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TotalPlant Solution (TPS) System

TPS System Audit Tool

User's Guide

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About This Document

This document describes the installation, preparation, and running of the System Audit Tool.

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References

The following list identifies all documents that may be sources of reference for material discussed in this publication.

Document Title
<i>Engineer's Reference Manual, SW09</i>
<i>TPS System Configuration Utility User's Guide, TP20X</i>
<i>File Transfer Installation and Use, FT05</i>

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Symbol Definitions

The following table lists those symbols used in this document to denote certain conditions.

Symbol	Definition
	ATTENTION: Identifies information that requires special consideration.
	TIP: Identifies advice or hints for the user, often in terms of performing a task.
	REFERENCE -EXTERNAL: Identifies an additional source of information outside of the bookset.
	REFERENCE - INTERNAL: Identifies an additional source of information within the bookset.
CAUTION	Indicates a situation which, if not avoided, may result in equipment or work (data) on the system being damaged or lost, or may result in the inability to properly operate the process.

Symbol Definitions

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1. Introduction

1.1 The System Audit Tool

The System Audit Tool performs analysis of data about the hardware and software found on a TPS process control network (the LCN, UCN, and data hiway, collectively known as the TPS Network or TPN) and a Microsoft Windows-based TPS network. The tool consists of the following two primary components:

- TPN System Audit overlay, or AUDIT, and
- System Audit Report Generator, or Sargent.

The tool also depends on several external components, available separately as part of the TPS software distribution.

This analysis produces a report that is an audit of the hardware and software configuration of TPN and TPS networks. The report can be useful when diagnosing system malfunctions, and also when upgrading or reconfiguring the systems.

1. Introduction

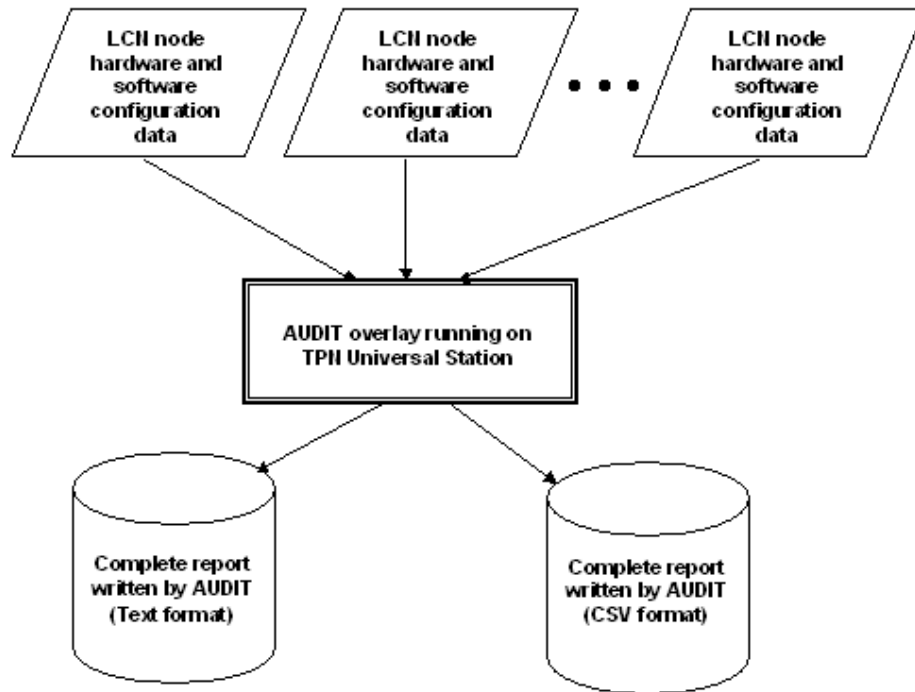
1.1. The System Audit Tool

AUDIT

AUDIT is a standard TPN overlay, available as part of recent releases of the GUS and US personalities. It is based on code taken from the LVRLOG tool, and its output is identical with that produced by LVRLOG. AUDIT reads data about TPN hardware and software configuration from the other nodes on its network, and produces two report files, one in human-readable text format (.txt), and the other in the common spreadsheet format known as comma-separated values, or CSV (.csv).

The operation of AUDIT is shown in Figure 1.

Figure 1 - Operation of AUDIT



Sargent

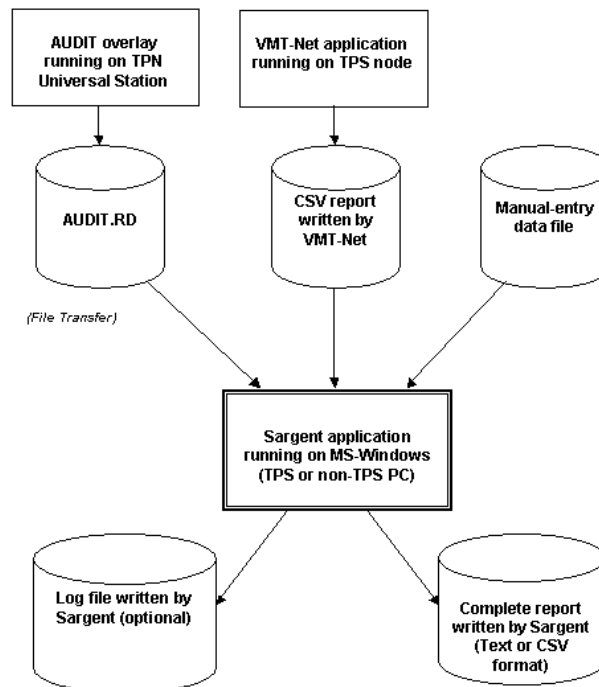
Sargent is a report generator that accepts the following inputs:

- A report generated by the AUDIT overlay. (Alternatively, reports from the LVRLOG tool may also be used by the current version of Sargent.)
- A report generated by the Networked TPS Version Management Tool –Net, VMT-Net.
- A text data file generated manually using a text editor.

Sargent processes and combines the input data, and produces a single report file containing both summary and detailed reports. It can produce its report in either text format (.txt), or in CSV format (.csv).

The operation of Sargent is shown in Figure 2.

Figure 2 - Operation of Sargent



1. Introduction

1.1. The System Audit Tool

External Components

In addition to AUDIT and Sargent, the System Audit Tool relies on two external software packages to collect and transfer the data used by Sargent itself. These external software components are categorized as “essential”, which are those components whose output is used directly by Sargent, and “recommended”, which are components which are useful to the operation of the tool, but for which alternatives exist.

Essential External Component

The essential external component of the System Audit Tool is VMT-Net.

VMT-Net

The VMT-Net application gathers information about hardware and software present on a TPS network. It must be installed and invoked separately from Sargent. Without VMT-Net input, Sargent will execute normally, but its report will not contain any data about the TPS network. It is fully documented in the VMT-Net section of the *TPS System Configuration Utility User's Guide*.

It can create output files in either HTML or CSV format, but only the CSV format is usable by Sargent. By default, VMT-Net names its output file according to the date and time during which it was created. This default name may be used as-is, or may be changed to any legal Microsoft Windows filename.

Recommended External Component

The recommended external component of the System Audit Tool is the TPS File Transfer application.

File Transfer

The File Transfer application permits the movement of files between a TPN HM and a Microsoft Windows storage location. It must be installed and invoked separately from Sargent. Its primary use is to move the AUDIT output file to a Microsoft Windows network where it will be accessible to Sargent. Without File Transfer the user will need to seek an alternate means for making the AUDIT data available to Sargent, if such data is to be used. File Transfer is fully documented in the *File Transfer Installation and Use* document.

1.2 Glossary

A glossary of the terms and abbreviations used in this document are contained in Table 1.

Table 1 - Glossary of Terms and Abbreviations

AUDIT	A standard US/GUS/UxS data collection overlay
CarbonCopy	A remote desktop control and sharing solution from Altiris Inc.
CSV	Comma Separated Value file format, a format for text data where fields are separated by commas
Data Hiway	Process control network connecting one or more TDC2000 controllers; connected to an LCN via a Hiway Gateway node
Desktop Streaming	A remote desktop control and sharing solution from Expertcity Inc.
FSC	FailSafe Controller
FSR	Field Services Representative
FTA	Field Termination Assembly
GUS	Global User Station
Hiway	See "Data Hiway"
Hiway Gateway	See "Data Hiway"
HG	Hiway Gateway
HM	History Module
HTML	Hypertext Markup Language
LCN	Local Control Network
LVRLOG	LCN Version/Revision Logging tool
NT	Windows NT
Sargent	System Audit Report GENeraTor, the Windows side report generator for the System Audit Tool
TPN	TotalPlant ® Network, also known as TPS Network
TPS	TotalPlant ® Solution

1. Introduction

1.2. Glossary

UCN	Universal Control Network
UNP	Universal Personality
US	Universal Station
UxS	Universal Station X
VMT	TPS Version Management Tool
VMT Net	Multiple TPS node version of VMT
W2000	Microsoft Windows 2000 [™] operating system
XP	Microsoft Windows XP [™] operating system

2. System Requirements

2.1 Hardware Requirements

The hardware requirements for running the System Audit Tool are very flexible. Since one primary purpose of the tool is to query the hardware installed on the networks being audited, it accepts any legal hardware configuration. The components do, however, require certain hardware components for their execution.

Table 2 - Hardware Requirements

System	Component	Description
TPN	AUDIT	The execution of the AUDIT component requires a Universal Station, a Universal Station (X), or a GUS. The hardware configuration of the other nodes connected to that US, UxS, or GUS has no restrictions; any legal TPN configuration is permitted.
TPS	VMT-Net	The execution of the VMT-Net component requires a PC running Microsoft Windows. This can be a TPS node, or a non-TPS Microsoft Windows machine connected via a network to the TPS nodes being audited. Refer to the VMT-Net documentation contained in the <i>TPS System Configuration Utility User's Guide</i> for more details.
	File Transfer	The File Transfer client component requires a PC running Microsoft Windows. To transfer files, it must be able to communicate with a PC where the File Transfer server is running. The server requires a PC with an LCNI board installed, connected to the LCN where the AUDIT results reside. Refer to the File Transfer Installation and Use document for detailed information.
	Sargent	The only hardware requirement to run Sargent itself is a Windows platform. This machine need not be connected to any process control network, nor does it need to have any special additional hardware installed. Any machine capable of running Microsoft Windows XP (with Service Pack 2) or Microsoft Windows 2000 (with Service Pack 4) can support the execution of Sargent.

2.2 Software Requirements

The software requirements for running the System Audit Tool are very flexible. Since one primary purpose of the tool is to query the software installed on the networks being audited, it accepts any legal software configuration. The components do, however, require certain software environments for their execution.

Table 3 - Software Requirements

System	Component	Description
TPN	AUDIT	The execution of the AUDIT component requires a Universal Station, a Universal Station (X), or a GUS loaded with the standard personality. The software configuration of the other nodes connected to that US, UxS, or GUS has no restrictions; any legal TPN configuration is permitted.
	VMT-Net	The execution of the VMT-Net component requires Microsoft Windows 2000 or XP. This can be running on a TPS node or a non-TPS Microsoft Windows machine connected via a network to the TPS nodes being audited.
	File Transfer	The execution of the File Transfer component requires Microsoft Windows 2000 or XP.
TPS	Sargent	The execution of Sargent requires Microsoft Windows 2000 or XP. This machine can be a TPS node, or a non-TPS Microsoft Windows machine.

3. Installation

3.1 Installing External Components

Installing VMT-Net

If you wish to include TPS data in your audit, you must install the VMT-Net application on every TPS node on each TPN network for which you wish to gather data. VMT-Net is supplied on the Honeywell TPS System Software CD, and uses an install wizard to guide you through the installation process. Refer to the *TPS System Configuration Utility Guide* for a detailed procedure for installing VMT-Net.

Refer to Table 4 for an overview of the VMT-Net installation procedure.

Table 4 - Procedure for Installing VMT-Net

Step	Action
1	Insert the TPS System Software CD into the CD-ROM drive. If the Honeywell CD navigation window does not appear automatically, then select the CD-ROM drive from Windows Explorer.
2	Select the "Install Packages" target on the navigation window.
3	When the Welcome dialog displays, read the details and if you agree, click Next .
4	Accept the software license agreement terms by clicking Next . Note the third party software information, and click Next .
5	If this dialog has been used before, the information will be filled in; if not, enter the Name, Company, License No., and Authorization No. information, and then click Next . RESULT: After the License No. and Authorization No. have been validated, the "Package Selection" dialog displays a list of the available licensed packages.
6	Select and install Version Management Tool

3. Installation

3.2. Installing AUDIT

Installing File Transfer

If you wish to include TPN data in your audit, you must transfer the AUDIT report from the TPN History Module where it is created to a Microsoft Windows disk where Sargent can read it. The simplest way to do this transfer is using the File Transfer application. Install the File Transfer application on at least one TPS node that is connected to the LCN where AUDIT is to be executed. It is installed in the same manner as all standard TPS software components; refer to the *File Transfer Installation and Use* document for installation instructions.

3.2 Installing AUDIT

The AUDIT overlay is a standard component of recent TPN software releases. It is a “generic overlay”, and can be executed either from its standard installation location on the HM (in **&OVG**) or from a TPN-formatted removable volume, or an emulated volume file on a GUS. In the latter case, use the US Engineering menu (**Support Utilities > Modify Volume Paths**) to set the volume path for generic overlays to indicate the location of the **audit.ov** file on the mounted volume.

3.3 Installing Sargent

Sargent is available on its own CD or on the TPS System Software CD.

The default installation location is **C:\Program Files\Honeywell\System Audit Tool**.

Use the appropriate installation procedure, depending on your CD.

Table 5 - Procedure for Installing Sargent – Sargent CD

Step	Action
1	If you received Sargent on a CD, insert the CD into a PC's CD-ROM drive. The installation wizard appears. If the installation wizard does not appear automatically, then select the CD-ROM drive from Windows Explorer. You should see the SETUP.EXE file. Execute that file, either by double-clicking the file in Windows Explorer, or by entering the pathname in the Start>Run text box.
2	Click Next. This displays the License Agreement page. Note: At any point during the installation process where a Back button appears, you may click that button to return to the previous step.
3	Review the license agreement and click Yes. This displays the Customer Information page.
4	Enter your user name and company name in the text boxes. Decide who should be allowed to execute Sargent on the installation machine (everyone or only the installing user) and select the appropriate option button. Finally, click Next. This displays the "Setup Type" page.

3. Installation

3.3. Installing Sargent

Step	Action
5	Choose the parts of the Sargent package to install: • Typical - installs the entire package. • Compact - installs only the executable program, and will not install the example files, or the Sargent User's Guide. • Custom - permits you to select the individual components of the package (executable, examples, documentation, and so forth.) It also allows you to choose a different destination location. Note: Typical is recommended unless the free disk space on the target machine is not adequate. In that case, select the Compact or Custom option to minimize the amount of space that is used.
6	After you choose your setup type, click Next. RESULT: • If you selected Custom, the 'Choose Destination Location' page displays. Go on to Step 7. • If you selected Typical or Compact, the 'Start Copying Files' page displays. Skip to Step 9.
7	If you selected Custom in Step 5, you can change the location where the package will be installed The default location is C:\Program Files\Honeywell\System Audit Tool To change the location, click Browse and browse to the new location. When the Destination folder displays the new location, click Next. This displays the 'Select Features' page.
8	On the 'Select Features' page, choose the parts to install. When you are satisfied with your selections, click Next. This displays the 'Start Copying Files' page. You may choose which parts of the overall package you wish to install, by checking and un-checking the checkboxes in the selection area. Beneath that area, the total disk space required for those parts is shown, along with the available disk space on the target volume.  ATTENTION If you do not install the "Program Files" component, you will not be able to execute Sargent.

Step	Action
9	On the 'Start Copying Files' page, click Next . This initiates the copying of the package files to their destinations, and will perform other updates to complete the installation. After the installation is complete, the InstallShield Wizard Complete page will appear.
10	Click Finish to terminate the wizard

Table 6 - Procedure for Installing Sargent – Sargent CD

Step	Action
1	Insert the TPS System Software CD .
2	Click Install Packages from the dialog.
3	When the Welcome dialog displays, read the details and if you agree, click Next .
4	Accept the software license agreement terms by clicking Next .
5	Note the third party software information, and click Next .
6	If this dialog has been used before, the information will be filled in; if not, enter the Name, Company, License No., and Authorization No. information, and then click Next . RESULT: After the License No. and Authorization No. have been validated, the "Package Selection" dialog displays a list of the available licensed packages.
7	Select System Audit Tool and click Install Package . Select OK button to install Sargent.

3. Installation

3.3. Installing Sargent

4. Preparation

Before running Sargent to produce the finished audit report, you must run AUDIT, VMT-Net, or both to create the input files that Sargent processes to create its own report.

Your choices are as follows:

- Run only AUDIT. This will allow Sargent to create a report containing only data about the TPN nodes on the LCN, UCN, and Data Hiway.
- Run only VMT-Net. This will allow Sargent to create a report containing only data about the TPS nodes.
- Run both AUDIT and VMT-Net. This will allow Sargent to create a report containing data about TPN and TPS nodes.

4. Preparation

4.1. Collecting TPN Data

4.1 Collecting TPN Data

The AUDIT overlay collects data about hardware and software on the TPN network. It is a standard overlay, and is invoked from the US (or Native window) command processor.

Using the AUDIT Overlay

The procedure for using the Audit overlay is provided in Table

Table 6 - Procedure for Using the AUDIT Overlay

Step	Action
1	Invoke the command processor on a US, GUS, or UxS.
2	Decide which option values and settings you want to use for this audit. Using the command-line syntax as described in <i>AUDIT Command Line Reference</i> section, type the appropriate AUDIT command. Figure 3 shows an example of an AUDIT command ready to be executed from the command processor.

Step	Action
------	--------

Figure 3 - AUDIT Command Ready for Report Generation

```
16 Jun 03 13:50:23 3
USER PATH : NET>TEST>

audit lcn dh ucn -s site_id -n lcn_id -o net:&ovg
```

Specifically, the command shown in Figure 3 will use the following values and settings:

- LCN nodes, Data Hiway boxes, and UCN nodes will all be audited.
 - This is the default if all of these keywords are omitted; they are shown above for illustration purposes.
- The site ID will be shown on the report as "SITE_ID".
- The LCN ID will be shown on the report as "LCN_ID".
- Headings will be included in the CSV file, since the "-h" option was not given.

The output will be written to **NET>&OVG>AUDIT.RD** and **NET>&OVG>AUDIT.RE**. Again, **NET>&OVG** is the default location if the "-o" option is omitted, and is included above for illustration purposes only

4. Preparation

4.1. Collecting TPN Data

Step	Action
3	Activate the overlay by pressing the ENTER key. Data collection and report generation begins. A series of progress messages should appear as shown in Figure 4. This process can take several minutes to complete.

Figure 4 - AUDIT Report Generation in Progress

```
16 Jun 03 13:55:40 3
USER PATH : NET>TEST>

AUDIT -S SITE_ID -N LCN_ID -O $F5>DATA
Starting audit report generation for LCN, Data Hiway, and UCN
Options: site ID: SITE_ID, LCN ID: LCN_ID
Options: output dir: $F5>DATA, headings: included
Auditing LCN nodes ...
Auditing LCN node 02
Auditing LCN node 06
Auditing LCN node 24
Auditing LCN node 25
Auditing LCN node 46
Auditing LCN node 54
Auditing LCN node 58
Auditing LCN node 60
Auditing LCN node 62
Auditing Data Hiway boxes ...
Auditing hiway 05 box 04
Auditing UCN nodes ...
audit -s site_id -n lcn_id -o $f5>data
```

Step	Action
	When the overlay has completed its execution, it will print one of two different messages: Report Generation Complete. (If the audit completed without error.) *** REPORT INCOMPLETE DUE TO FATAL ERROR *** (If errors were detected during the audit.) At this point, the report is ready. It will be available in two files, each in a different format as described below: • &OVG>AUDIT.RD - Human-readable text. This file can be viewed on the US (or Native Window) using the TPN text editor command "ed". If you choose to view the report in this manner, it is recommended that you use the F1 key to abort the edit, rather than re-saving the file and thereby taking a risk of modifying it. It may also be printed using the TPN "pr" command. This file may be used as input by the Sargent application, but it is recommended that you use the CSV-format AUDIT.RE when possible. • &OVG>AUDIT.RE - Machine-readable CSV file. This file may be used as input by the Sargent application.
	ATTENTION &OVG is the default location for the AUDIT output files, and it is used here for purposes of illustration. The files may actually be put in any writeable location on the TPN network via the "-o" command-line option.

4. Preparation

4.1. Collecting TPN Data

AUDIT Command-Line Reference

Command Syntax

The syntax of the AUDIT command line is as follows:

```
audit [ keyword-or-option ] ...
```

Where *keyword-or-option* may be zero or more of the following:

Keyword	Definition
lcn	audit LCN nodes
dh	audit Data Hiway nodes
ucn	audit UCN nodes
Option	Definition
-s <i>site-id</i>	set the site ID on the report to <i>site-id</i>
-n <i>lcn-id</i>	set the LCN ID on the report to <i>lcn-id</i>
-h	omit report headings
-o <i>out-path</i>	set the output path (dev>dir) to "out-path"
-?	print the overlay's help message

This information is also shown on the AUDIT help display as shown in Figure 5.

Figure 5 – AUDIT Help Display

```
16 Jun 03 13:43:52 3
USER PATH : NET>TEST>

AUDIT -?

Syntax:

    audit [ keyword-or-option ] ...

Where "keyword-or-option" may be zero or more of the following:

Keywords:
    lcn          audit LCN nodes
    dh           audit Data Hiway nodes
    ucn          audit UCN nodes

Options:
    -s site-id   set the site ID on the report to "site-id"
    -n lcn-id    set the LCN ID on the report to "lcn-id"
    -h           omit report headings
    -o out-path  set the output path (dev>dir) to "out-path"
    -?          print this help message

If no keywords are given, each of the LCN, Data Hiway, and UCN are audited.
The default site-id and lcn-id are blank. Headings are included by default.
The default out-path is "NET>&OVC".
```

Syntax Notes

- If *none* of the keywords (**lcn**, **dh**, or **ucn**) are given, each node on the LCN, Data Hiway, and UCN will be audited. If one or more keywords are specified, then only those listed will be audited. Omitting the keywords entirely is the same as specifying all of them.
- The default *site-id* and *lcn-id* values are blank. Each value may be from 1 to 8 characters in length, and may include letters, numbers, and all punctuation characters except quote marks (" ").
- Headings are included by default. If the **-h** option is specified, secondary headings will be omitted from the CSV version of the report. This makes the AUDIT report more easily used in a spreadsheet. The human-readable (.txt) report is not affected by

4. Preparation

4.1. Collecting TPN Data

this option. The Sargent report generator ignores these header lines if they are present, so the **-h** option has no effect on reports produced by Sargent.

- The default *out-path* is **NET>&OVG**. Any location that is visible and writeable from the US (or Native Window) where AUDIT is being run, may be specified. The format can be either of the following:

- `dir`
- `dev>dir`

A trailing “>” is optional (so `dir>` and `dev>dir>` are also acceptable. The *dir* portion is a valid directory name from 1 to 4 characters in length, and the optional *dev* portion is a valid device name from 1 to 6 characters in length. If the *dev* portion is omitted, the volume name **NET** is used. Valid examples are **DATA**, **ADIR>**, **NET>TEMP**, **\$F12>DATA>**, and **\$F2>RPT**.

- The help message is printed in response to the **-?** option, or when a command-line syntax error is detected. In either case, no auditing is performed. Any output files which may already have been present in the output directory are unaffected. The help screen is shown in Figure 5
- Alphabetic case is not significant on the command line. The US (or Native Window) command processor forces all command lines to upper-case before processing them, so any case you use will be treated as upper-case.
- If more than one value-setting option (**-s**, **-n**, or **-o**) with the same name is given, the rightmost (“later”) value is used. So “**-s foo -s bar**” would set the site ID value to “bar”.
- Any values, such as *site-id*, *lcn-id*, or *out-path*, which are longer than the size limits given above, will be truncated at their maximum length, and that value will be used instead.
- In an attempt to increase usability, certain flexibility is allowed in the command-line syntax:
 - Additional spaces (more than one) after the command name, and between keywords, options, and values, are not significant.
 - Spaces between value-setting options (**-s**, **-n**, or **-o**) and their associated values are optional. Thus, “**-s foo**” and “**-sfoo**” have the same effect.
 - Only the first letter of each keyword is significant, so for example, “**L**”, “**LC**”, and “**LCN**” all request an audit of LCN nodes, as do “**LCNNODES**” and “**LARRY**”.

- Keywords may have a dash before them, like options. Thus, “ucn” and “-ucn” (or “d” and “-d”) have the same effect. The same is true of the “-?” option, so “?” by itself is also recognized as a help request.
- The keywords and options may be specified in any order.

Default AUDIT Operation

Given the defaults, executing the command **audit** with no options will perform the following audit:

- All LCN nodes, Data Hiway boxes, and UCN nodes will be audited.
- The site ID and LCN ID fields on the report will be blank.
- Secondary headers will be included in the CSV report. As noted above, this has no effect on Sargent.
- The output files will be written to:

NET>&OVG>AUDIT.RD and **NET>&OVG>AUDIT.RE**.

4. Preparation

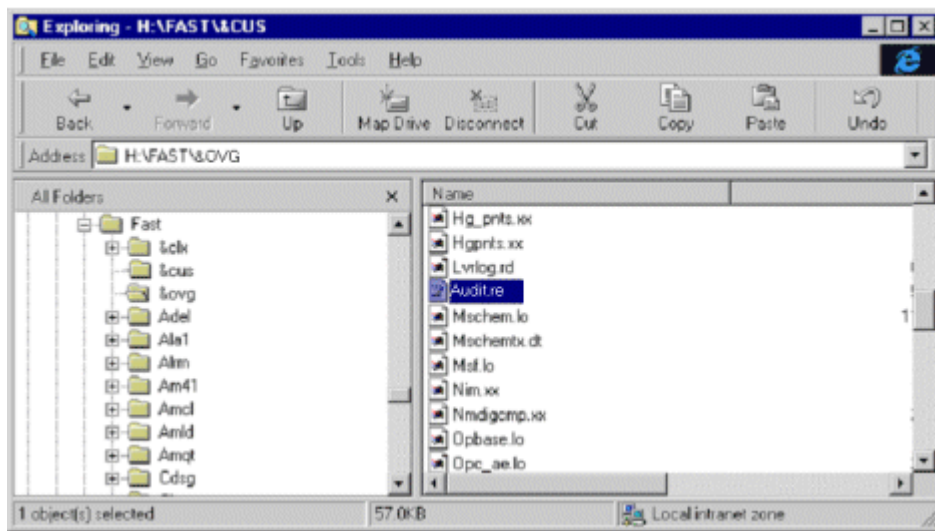
4.1. Collecting TPN Data

Making TPN Data Available to Sargent

Once AUDIT has successfully created its report files, you now need to make at least one of the report files available for Sargent to use. It is recommended that the CSV file, (**AUDIT.RE**) be available for Sargent. This is done using the File Transfer application.

File Transfer makes TPN History Modules available to TPS nodes. They appear as Microsoft Windows network drives, and are available to be mapped to a drive letter just as with any other shared drive on the Microsoft Windows network. Refer to the *File Transfer Installation and Use* document for complete details on how to set up that application. Figure 6 shows a Windows Explorer with an HM mapped to drive H using File Transfer. The **AUDIT.RE** file is highlighted in the right-hand pane of Explorer.

Figure 6 - Windows Explorer Showing AUDIT Report Selected



After File Transfer has been installed and configured, a connection made to the TPN network, and the History Module containing the AUDIT data has been mapped to a drive letter, you have two options for making the AUDIT data available to Sargent: directly, or by copying the file.

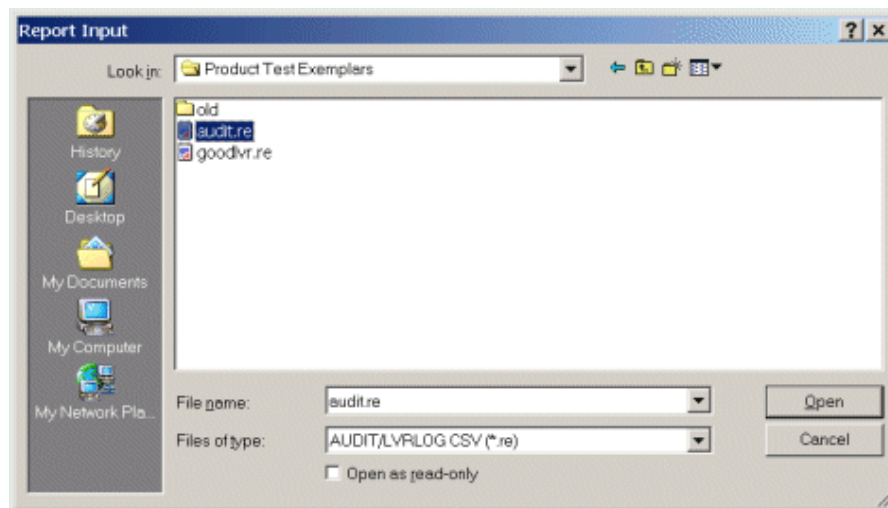
Using AUDIT Data Directly

One way of making the AUDIT data available is to instruct Sargent to read it directly from the History Module. This can be accomplished as described in the Table 7.

Table 7 - Procedure for Using AUDIT Directly

Step	Action
1	Ensure that the History Module is mapped to a drive letter as shown in Figure 6.
2	Select the AUDIT/LVRLOG file checkbox in the Sargent window. This enables the Browse button to the right of the filename text box.
3	Click the Browse button. This causes the file browser dialog window to open. An example of Sargent's file browser dialog is shown in Figure 7

Figure 7 – Browsing to the AUDIT Input File



- Using the file browser, navigate down the mapped drive's directories to the **&OVG** directory.

This contains the **AUDIT.RE** file, unless the **-o** option was used to direct the output to a different directory.

4. Preparation

4.1. Collecting TPN Data

Step	Action
5	Select the AUDIT.RE file. The filename will be highlighted
6	Click the Open button at the lower right. This causes the complete pathname to appear in the AUDIT/LVRLOG file text box on the Sargent window. In Figure 6, the complete pathname of the AUDIT.RE file is H:\FAST\&OVG\AUDIT.RE . When Sargent is instructed to generate its report, it will read the AUDIT data directly from the HM.

As an alternative to using the file browser to locate the **AUDIT** data file, if you already know the complete pathname of the file, you may enter it directly into the **AUDIT/LVRLOG file** text box on the Sargent window.

The advantage of reading the **AUDIT** data directly from the HM is that it eliminates the initial step of copying the file to an Microsoft Windows disk, as described in the *Copying AUDIT Data to Microsoft Windows* section. The disadvantage is that being a networked connection; the reading of the file takes somewhat more time than reading it directly from a local disk drive.

Copying AUDIT Data to Microsoft Windows

Another way to make the AUDIT data available to Sargent is to use File Transfer to copy it to an Microsoft Windows disk drive before directing Sargent to read the data. This can be accomplished using the procedure in Table 8.

Table 8 - Procedure for Copying AUDIT Data to Microsoft Windows

Step	Action
1	Open Windows Explorer.
2	Ensure that the History Module is mapped to a drive letter. Figure 6 shows the HM mapped to drive H .
3	Using Explorer, navigate down the mapped drive's directories to the &OVG directory. This contains the AUDIT.RE file, unless the -o option was used to direct the output to a different directory.
4	Select the AUDIT.RE file and drag the filename to a folder location on the Microsoft Windows network. This can be somewhere on the C drive, a removable drive, another mapped drive on the Windows network, or anywhere that you have permission to write a file. Release the mouse button to initiate copying of the file from the HM to the new location. Note the file location, because Sargent will need to know the location.
5	Select the AUDIT/LVRLOG file checkbox in the Sargent window. This enables the Browse button to the right of the filename text box.
6	Click the Browse button. This causes the file browser dialog window to open. An example of Sargent's file browser dialog is shown in Figure 7.
7	Using the file browser, navigate to the location to which you copied the AUDIT.RE file in Step 4.
8	Select the AUDIT.RE file. The filename will be highlighted, and will appear in the File name text box.
9	Click the Open button at the lower right. This causes the complete pathname to appear in the AUDIT/LVRLOG file text box on the Sargent window. When Sargent is instructed to generate its report, it reads the AUDIT data from this location.

4. Preparation

4.1. Collecting TPN Data

As an alternative to using the file browser to locate the `AUDIT` data file, you may enter the complete pathname of the copy directly into the **AUDIT/LVRLOG file** text box on the Sargent window.

The advantage of reading the `AUDIT` data from a local copy is that it is somewhat faster than reading it directly from the HM, and the pathname may be somewhat easier to remember. If you wish to run Sargent again, to create a report in a different format, or to use a different set of manual-entry data, for example, then having a local copy can be convenient. Also, the file may be copied elsewhere for example, to removable media for additional processing by other applications. The disadvantage is that you must actually make the copy first.

4.2 Collecting TPS Data

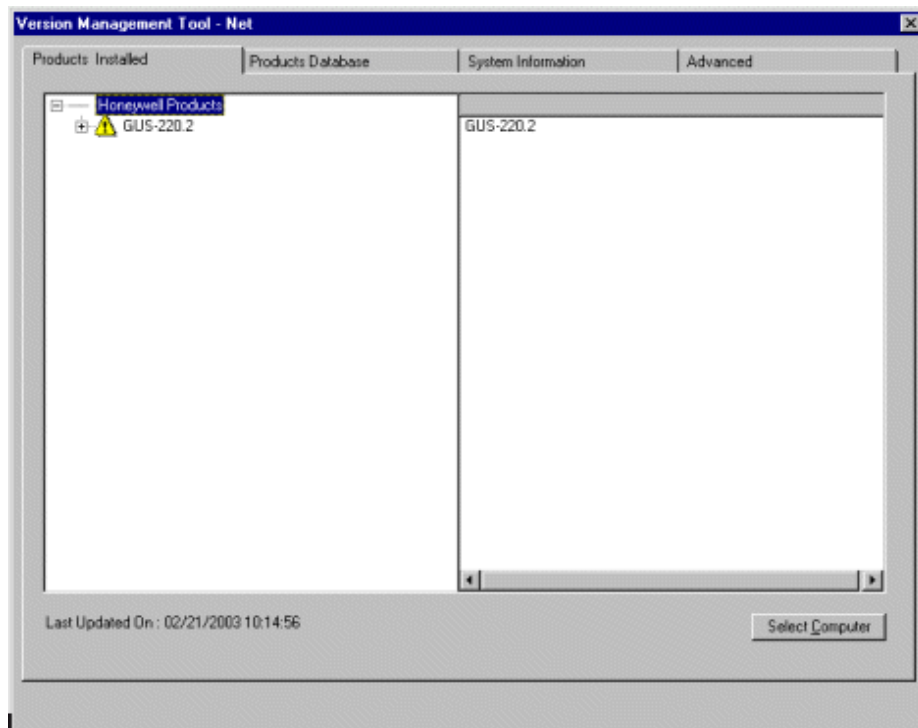
Using VMT-Net Tool

Perform the following procedure to collect TPS data using the VMT-Net tool.

Table 9 - Procedure for Using the VMT-Net Tool

Step	Action
1	Invoke the VMT-Net application, either from the Programs > Honeywell TPS menu, or by using the shortcut that the installation created on the desktop. The initial VMT-Net display appears as shown in Figure 8.

Figure 8 – Initial Display of VMT-Net Tool

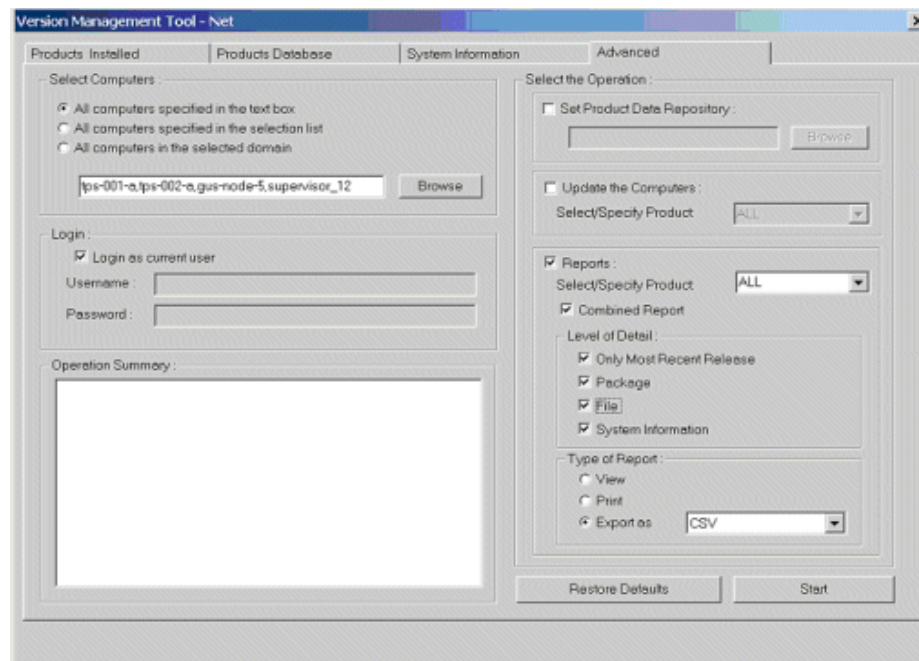


4. Preparation

4.2. Collecting TPS Data

Step	Action
2	Select the Advanced tab at the top of the VMT-Net window. The Advanced tab appears as shown in Figure 9.

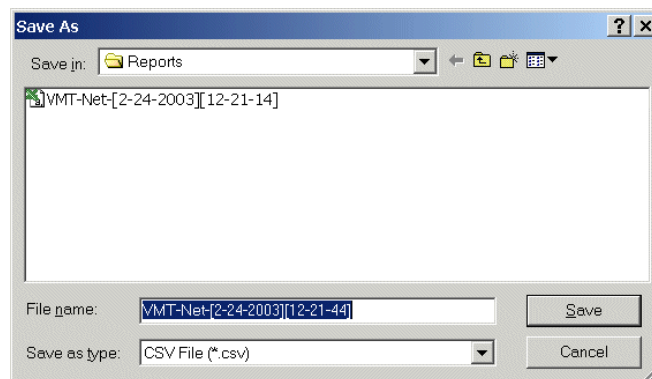
Figure 9 – Advanced Page – VMT-Net Tool Display



- 3 Select the nodes to be audited in the **Select Computers** area at the upper left, as described in the VMT-Net manual.
 - 4 Ensure that the following settings are in effect in the area at the lower right of the window as shown on Figure 9, starting with the **Reports** checkbox:
 - **Reports** is selected.
 - **Select/Specify Product** is set to **ALL**.
 - **Combined Report** is selected.
-

Step	Action
	Decide whether you wish to scan for all releases, or only the most recent release. Set the Only Most Recent Release checkbox appropriately in the Level of Detail area. It is recommended that selecting the checkbox and scanning only for the most recent release. If you do not check the Only Most Recent Release box, the resulting report can be very large. Ensure that you make the following selections: • Package is selected. • File is selected (not selected by default, so you will usually need to select it). • System Information is selected. • Export as option is selected. • CSV format is selected (by default, VMT-Net generates its report in HMTL format, so you will usually need to select the pull-down and change it to "CSV").
5	Click the Start button at the lower right. A pop-up dialog will appear to confirm that the operation is going to be performed on the selected number of computers. Confirming the "number of computers" dialog will begin data collection and report generation. Progress messages will begin to scroll in the "Operation Summary" area at the lower left of the window. This process can take several minutes to complete.
6	Wait until the Save As dialog (Figure 10) pops up. This gives you the opportunity to set the name of the report file, and the name of the directory in which to save it.

Figure 10 - VMT-Net Save-Report Dialog



4. Preparation

4.2. Collecting TPS Data

Step	Action
7	Select an appropriate file name and folder in which to save the report. The defaults are acceptable, but you should record the name and location of the report file for later use as input to Sargent. By default, VMT-Net places its reports in \\Program Files\\Honeywell\\Tps\\VMTNet\\Reports The name of the file is created using the date and time-of-day of the report generation. The following file name is shown in the example in Figure 10. VMT-Net-[2-24-2003][12-21-44]
8	You can now close the VMT-Net application, because continued execution is not needed by Sargent.

4.3 Manual-Entry Data File

Some of the data that the System Audit Tool uses to create its report cannot be gathered programmatically, and thus must be entered manually. Using the format described below, place any available system-audit data into this file, and when you invoke Sargent, you can provide the pathname of this auxiliary-data text file. If the file is not present, is empty, or is not readable, no manually entered data will be included in Sargent's report.

The manual entry data file can be created and maintained either on the TPN side or on the Windows side. Sargent accesses the manual entry data file from the Windows file system.

If the log file is enabled (refer to Section 5.3, *Specifying Output and Log Files*) Sargent will log any errors encountered during processing of the manually entered data file.

Creating and Maintaining

The manual-entry data file is a standard flat-text file. You may create it using either the standard TPN system text editor available at any US, or any Microsoft Windows text editor. If you use some other editing application, such as Microsoft Word, ensure that you save the file in plain text format. In Word, this is done by selecting the **Save** as type pulldown on the **File>Save** dialog, and choosing the format: Microsoft DOS Text with Line Breaks. The file may be maintained either on the LCN on a History Module, or in an Microsoft Windows folder.

There are advantages to each approach as follows:

- Maintaining the file on the LCN keeps it “close” to the hardware and software it is describing, since currently the directives it can contain describe only LCN components. Also, HM's at most sites tend to be a less “active” environment than most Microsoft Windows disks, so the chances of the file being deleted or overwritten are often lower. However, the text editor on the US is considered to be more difficult to use than those available on Microsoft Windows. Also, if the file is maintained on an HM, it must be made available to Sargent in the same way as the AUDIT data as described in the *Making TPN Data Available to Sargent* section. The chance of using an out-of-date version of the file (if a previous file exists on an Microsoft Windows disk from an earlier audit session) also increases if the file is maintained on the LCN.
- Maintaining the file on Microsoft Windows means that it is available for Sargent to read, and can be maintained by more powerful tools. Locating the file during subsequent sessions may be more difficult, however, and safeguarding the data it contains may require more effort.

4. Preparation

4.3. Manual-Entry Data File

The preferences of the user and the standard practices at a given site determine how and where the manual-entry data file is maintained.

The documentation for the Sargent package includes a template file for the manual-entry data. This file contains comments that describe the format of the data directives and the meaning of their fields, and which give examples of properly-formed instances of each directive. This template file may be copied to a different name, and used as a starting point for constructing the actual manual-entry data file for a given site, by opening it in a text editor (such as MS Notepad) and by copying and modifying the example directives to contain the correct data for the network being audited.

File Format

The manual-entry data file uses a simple leading-keyword format. Each line in the text file starts with a unique keyword identifying the data that line contains. The format of the rest of the line is determined by the keyword, but all lines consist of tokens separated by spaces.

The number and location of spaces does not matter to the AUDIT tool. Alphabetic case does not affect recognition of the keywords, so “fta”, “Fta”, and “FTA” are all valid, and are all considered the same keyword. Blank lines are ignored; lines beginning with “&” or “# ” are considered to be comments, and are also ignored. Thus, the file may be formatted and annotated to provide additional information to future users, and to make it easier to read and update.

All data directives listed in the *Data Directives* section are optional. Any lines that are badly formed and unrecognizable will be reported as errors, and their contents will not appear in the report. All valid directives present in the file will be formatted and will appear in the report, along with a visual indication that the data was manually entered rather than gathered directly from the system by the tool itself. The column headers of manually-entered data items have an asterisk (“*”) after the column name. As an example, refer to the FTA* or REV* columns in the sample output in Section 7.

Data Directives

Box-Card

- **box-card** hiway-number box-number card-number card-type revision1 [revision2]*

* “revision2” is an optional field; cards may have either one or two associated revision fields

This directive provides a way to add card information for a card within a DHP (or other hiway boxes for which card information is not available). The user can specify the card number, card type, revision1 and revision2 associated with the card. The maximum total number of cards, including those added using this directive and those already listed as part of the hiway box, is 32. The data will appear on the Hiway Detail Report and the Board List.

The *hiway-number* and *box-number* fields are used by Sargent to locate the correct Hiway box in the AUDIT data. The *card-number*, *card-type*, *revision1* and optional *revision2* fields provide the actual card configuration data. A physical card with the number *card-number* in that box must **not** be detected by AUDIT. The fields have the following definitions:

- **hiway-number:** the hiway number with which the DHP or other box type is associated; integer 1 to 20
- **box-number:** the box number of the DHP or other box type; integer 1 to 63
- **card-number:** the card number of this card within the box; integer 1 to 32
- **card-type:** the card type of this card; string 1 to 9 characters
- **revision1:** the first revision number associated with this card; string 1 to 4 characters
- **revision2:** if the I/O card has two revision numbers associated with it, this optional field provides the second revision number; string 1 to 4 characters

4. Preparation

4.3. Manual-Entry Data File

Box-Rev

- **box-rev** hiway-number box-number card-number revision1 [revision2]*

* “*revision2*” is an optional field; boxes may have either one or two associated revision fields

This directive provides the firmware revision(s) for an I/O card within a controller box on the Data Hiway. Some boxes have only one revision number, and some have two.

The *hiway-number*, *box-number*, and *card-number* fields are used by Sargent to locate the correct I/O card in the AUDIT data. The *revision1* and optional *revision2* fields provide the actual revision number data. The fields have the following definitions:

- **hiway-number:** the number of the Data Hiway on which the box is located. This data may be found on the US (or Native Window) System Status display, or in the AUDIT report. Integer 1 to 20.
- **box-number:** the number of the controller box containing the specified I/O card. This data may be found on the US (or Native Window) Network/Hiway Status display for the selected Data Hiway, or in the AUDIT report. Integer 1 to 63.
- **card-number:** the number of the I/O card for which the revision number(s) are being provided. This data may be found via physical inspection of the controller box, or in the AUDIT report. Integer 1 to 32.
- **revision1:** provides the first (or only) revision number for the I/O card. String 1 to 4 characters.
- **revision2:** if the I/O card has two revision numbers associated with it, this optional field provides the second revision number. String 1 to 4 characters.

Fta

- **fta** ucn-number node-number iop-number fta-type version

This directive provides the type (model number) and firmware version of the Field Termination Assembly that connects an I/O processor to the actual process hardware. One FTA can serve either a single IOP, or both members of a redundant pair.

The *ucn-number*, *node-number*, and *iop-number* fields are used by Sargent to locate the correct IOP(s) in the AUDIT data. The *fta-type* and *version* fields provide the actual data about the FTA. The fields have the following definitions:

- **ucn-number**: the number of the UCN on which the connected IOP is located. This data may be found on the US (or Native Window) System Status display, or in the AUDIT report. Integer 1 to 20.
- **node-number**: the number of the UCN node in which the connected IOP is installed. If the node is a member of a redundant pair, this must be the number of the **primary** member of the pair. This data may be found on the US (or Native Window) Network/Hiway Status display for the selected UCN, or in the AUDIT report. On that display, the primary's node number is the one appearing to the left of the node type. In the AUDIT report, IOPs that are assigned to a redundant pair have "PP/SS" in the "NODEN" column; if this value is, for example, "09/10", then the *node-number* field in the **fta** directive must be "09". The primary is always the lower-numbered of the two nodes. Integer 1 to 64.
- **iop-number**: the "module number" of the IOP which is connected to the FTA. In the case of a redundant pair of IOPs, this can be the module number of either the primary or the secondary IOP. This data may be found on the US Detail Status display for the selected UCN node. Note that PMs do not have redundant IOPs; only APMs and HPMS currently have this capability. Integer 1 to 64.
- **fta-type**: provides the type / model number of the FTA. This data may be found via physical inspection of the FTA. String 1 to 10 characters.
- **version**: provides the firmware version in use on the FTA. This data may be found via physical inspection of the FTA. String 1 to 4 characters.

Use one FTA directive per physical FTA. The type and version information will be added to the report lines for both the primary and secondary IOPs if the FTA is connected to a redundant pair.

4. Preparation

4.3. Manual-Entry Data File

Tpn-Rev

- **tpn-rev** *version*

This directive provides a way to associate a TPN release version with the TPN data reported by Sargent. The version will appear on the LCN Summary Report.

- **version:** the TPN software release version in a “human-readable” format; e.g. “R650.3” String 1 to 10 characters.

Ucn-Card

- **ucn-card** *ucn-number node-number card-number card-type software-version software-revision*

This directive allows the user to add card information for a card associated with a UCN node, such as an SM – FSC or SM – TRICON. The user can specify the card number, card type, software version, and software revision associated with the card. The maximum total number of cards, including those added using this directive and those already listed as part of the UCN node, is 64. The new data will appear in the UCN Detail Report and the Board List. It is especially useful for adding an I/O card to an FSC node.

The *ucn-number* and *node-number* fields are used by Sargent to locate the correct UCN node in the AUDIT data. The *card-number*, *card-type*, *software-version*, and *software-revision* fields provide the actual data about the card. The fields have the following definitions:

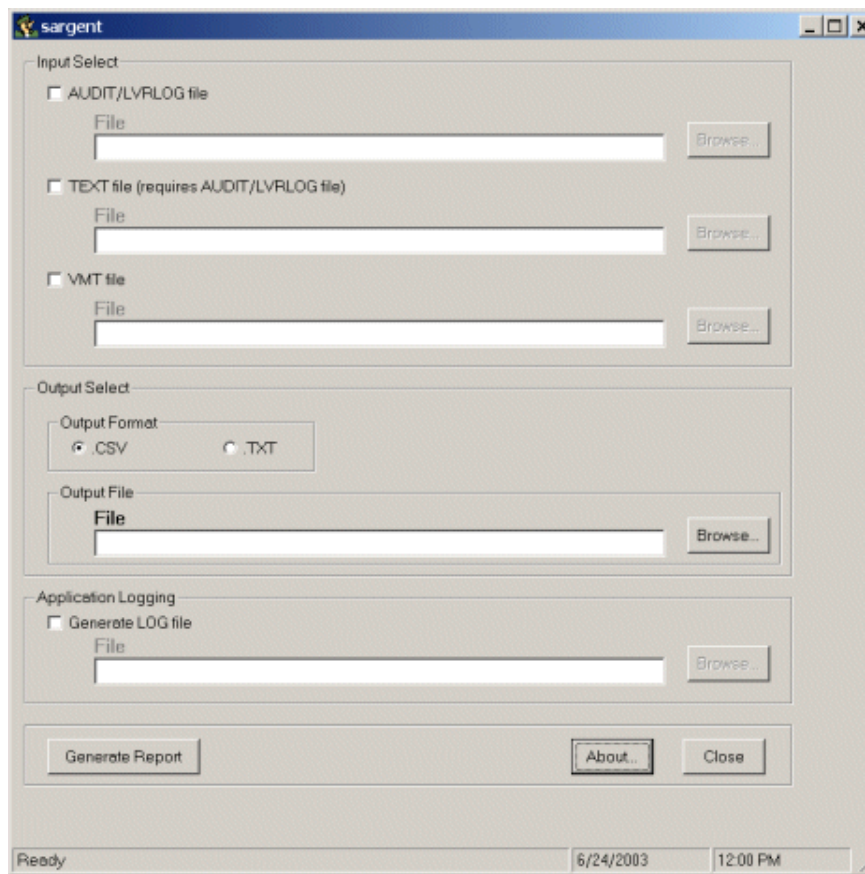
- **ucn-number:** the UCN number with which the FSC or other node type is associated. Integer 1 to 20.
- **node-number:** the node number of the FSC or other node type with which the card is associated. Where a card is associated with a redundant pair of nodes, *node-number* designates the primary. Integer 1 to 64.
- **card-number:** the card number of the card with which the type and version/revision data is to be associated; integer 1 to 64
- **card-type:** the card type; string 1 to 9 characters.
- **software-version:** the software version associated with this card; string 1 to 4 characters
- **software-revision:** the software revision associated with this card; string 1 to 4 characters

5. Running Sargent

5.1 Invoking Sargent

Sargent is invoked by selecting its menu pick from the Microsoft Windows Start menu. It is normally located directly on the Programs menu, as a selection named **Sargent**. Selecting this will invoke Sargent, which will display its initial screen configuration, as shown in Figure 11:

Figure 11 - Initial Display of Sargent

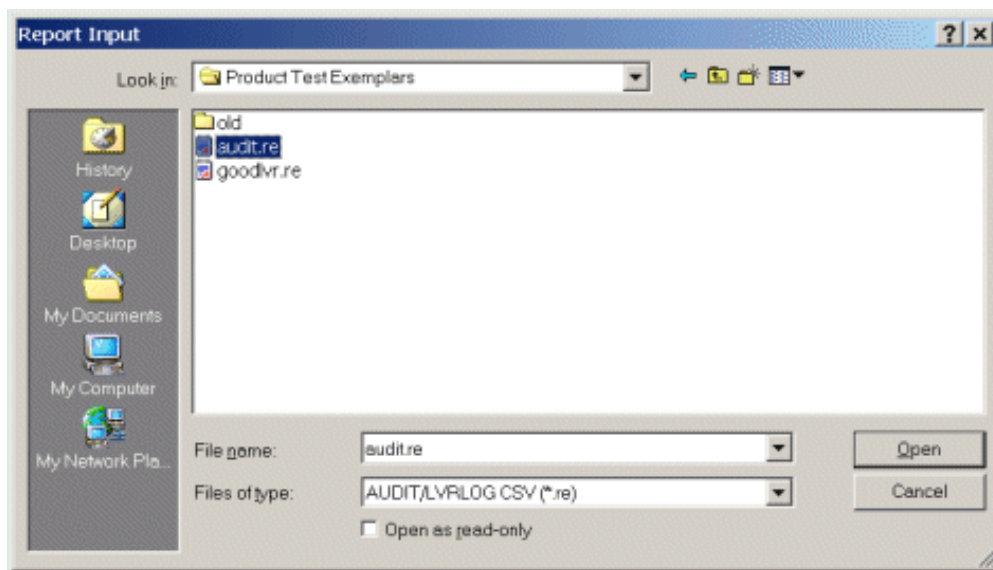


5.2 Selecting Input Files

Initially, the text boxes and **Browse** buttons associated with the three types of input files (AUDIT/ LVRLOG, TEXT, and VMT) are disabled. To enable them, select the associated checkbox that appears above and to the left of the associated text box shown in Figure 11. You then have the choice of entering a path into the input file directly into the text box, or selecting input files by browsing to them using the associated **Browse** command button.

If you select one of the Browse buttons, a file browser dialog will pop up, similar to the one shown in Figure 12.

Figure 12 - Browsing to the AUDIT Input File



For Sargent to generate an output file, at least one type of input file must be selected. Selection of the manual entry input file also requires the use of a corresponding AUDIT input file. This is necessary in order to validate the entries in the manual entry data file. Otherwise, selecting any particular type of input file is optional.

The following guidelines apply to the three input file entries:

- For the AUDIT input file, Sargent expects either an AUDIT CSV file (*.re) or an AUDIT text file (*.rd). The current version of Sargent will also accept LVRLOG CSV or text files.
- For the TEXT input file, also known as the manual-entry data file, Sargent expects a text file with the “.txt” extension. Each line of the text file must conform to the syntax set forth in Section 4.3- *Manual Entry Data File*.
- For the VMT input file, Sargent expects a CSV file with the “.csv” extension produced by VMT-Net. Sargent will validate that the input file was produced by VMT-Net.

5.3 Specifying Output and Log Files

Output File

Sargent requires a report output file path to generate reports. Select a report output file path either by entering it directly into the Output File text box, or by browsing to it using the associated **Browse** command button. The **Browse** button will invoke a pop-up file browser similar to the one shown in Figure 12. The difference in its use for output files (such as the report file and the log file) is that, instead of selecting an already-existing file, you should type a *new* name to be used for the file that Sargent will create. If you select the name of an existing file, the existing file will be overwritten.

Sargent can generate its report either in human-readable text format, suitable for browsing or printing, or in CSV format, readable by various external applications, such as spreadsheet and database programs. Choose one of the two radio buttons in the Output Format area on the Sargent window to direct Sargent to produce its report in the selected format.

Application Log File

If the **Generate LOG File** checkbox is selected, Sargent will generate a log file during report generation. It will log input and output file paths, the progress of report generation, and any file errors, as well as syntax errors encountered during processing of the manual-entry data file.

If a file error occurs during generation of the log file, a pop-up message appears describing the error, and application logging is turned off, to prevent the appearance of a cascade of subsequent error messages.

By default, the log file will be placed in the same folder as that selected for the output file. The default file name for the log file is **sargent.log**.

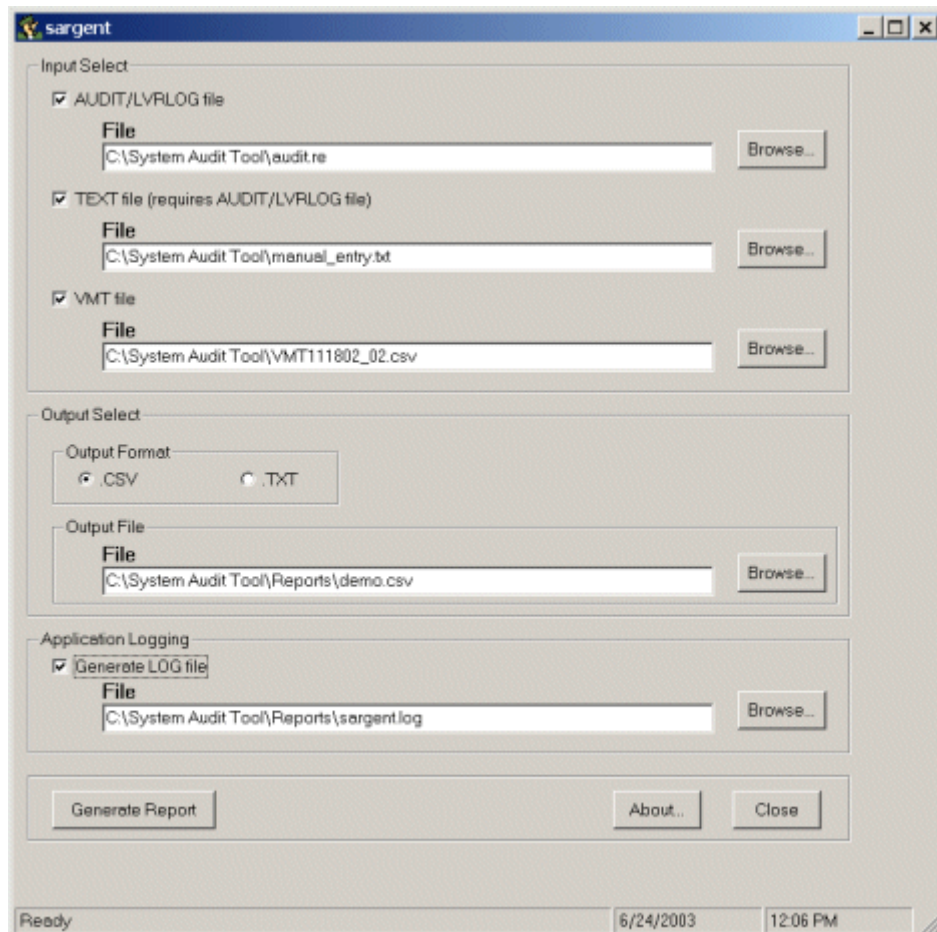
A different path may be specified in the same manner as for the output file path: Select a log file path either by entering it directly into the **Application Logging** text box, or by browsing to it using the associated **Browse** command button. The **Browse** button will invoke a pop-up file browser similar to the one shown in Figure 12. The log file pathname **must not** be the same as the output file's pathname.

To restore the default log file pathname once it has been altered, you may uncheck the **Generate LOG File** checkbox and check it again.

5.4 Generating Reports

Figure 13 shows a Sargent window with all options and entries selected. This would be typical for generating a “full” audit report, including TPN, TPS, and manual-entry data.

Figure 13 - Sargent Display Ready to Generate Reports



When one or more input files have been selected, an output file pathname has been specified, and (optionally) a log file pathname has been specified, Sargent is ready to generate its report. To begin report generation, select the **Generate Report** button at the lower left of the Sargent window shown in Figure 13.

5. Running Sargent

5.5. Exiting Sargent

The configuration in shown in Figure 13 causes Sargent to perform the following actions when the **Generate Report** button is selected:

1. Open the file **C:\System Audit Tool\Reports\sargent.log** and begin writing application event messages into it, and any error messages that may be required.
2. Open the file **C:\System Audit Tool\Reports\demo.csv** and begin writing report lines in CSV format into it.
3. Open the file **C:\System Audit Tool\audit.re** and read AUDIT data from it.
4. Open the file **C:\System Audit Tool>manual_entry.txt** and read manual-entry data from it.
5. Associate the manual-entry data with the appropriate AUDIT data items.
6. Open the file **C:\System Audit Tool\VMT111802_02.csv** and read VMT-Net data from it.

5.5 Exiting Sargent

When you are finished using Sargent, you may exit it in one of the following two ways:

- Select the **Close** button in the lower right of the Sargent window.
- Select the Microsoft Windows default close button (the “X”) in the Sargent window’s title bar.

In either case, a pop-up dialog will confirm the action by asking **Are you sure you want to Close?** Click the **Yes** button to terminate Sargent, or the **No** button to continue the execution.

5.6 Sargent Status Line

The Sargent status line appears as the bottom line of the Sargent display window as shown in Figure 13. The left panel shows a status of **Ready** at all times except during report generation, during which it reports on the progress of report processing. After the status returns to **Ready**, report generation is complete. The middle and right panes show the current system date and time, respectively.

5.7 Invoking the About Box

You may view a pop-up window displaying information about Sargent, including its version number, by clicking the **About** button in the lower right of the Sargent window shown in figure 13. The information window, or the About box, looks like the one shown in Figure 14.

Figure 14 – About Sargent Box



5. Running Sargent

5.7. Invoking the About Box

6. Execution Scenarios for System Audit Tool

6.1 “Walk-Up” TPN Report

If the TPN network to be audited has not yet been upgraded to a release that includes the AUDIT overlay as part of the standard load, perform the procedure in Table 10.

Table 10 - Procedure for Upgrading a TPN Release to include AUDIT Overlay

Step	Action
1	Obtain a version of the AUDIT overlay (audit.ov) that is compatible with the version of TPN software installed on the network to be audited. If you do not have the correct version of the AUDIT overlay, contact your Honeywell Technical Assistance Center. Any such versions that are available will be found at the following Solution Support Online web site: http://www.acs.honeywell.com/
2	Place a copy of the compatible audit.ov file on a TPN-formatted removable media volume, such as a floppy, a Bernoulli cartridge, or a Zip cartridge, which is readable by a US on the network to be audited. For a GUS, this can be a mountable GUS emulated volume file.
3	Insert the volume into a drive on a selected US.
4	From the Engineering main menu, select Support Utilities .
5	From the Support Utilities menu, select Modify Volume Paths .
6	Modify the Generic Overlay pathname (located at the lower-right of the volume-paths screen) to indicate the drive and directory where the audit.ov file is now located, on the removable volume. Press ENTER to accept the change.
7	Continue with Step 1 of <i>TPN-Only Report (Table 11)</i> or <i>TPN and TPS Report (Table 13)</i> .

6.2 TPN-Only Report

If only the TPN network is to be audited, perform the procedure in Table 11.

Table 11 - Procedure for Auditing a TPN Network

Step	Action
1	Ensure that AUDIT is installed on a US or UxS on the TPN network, File Transfer is installed to make the AUDIT data available to Sargent, and that Sargent is installed on an Microsoft Windows machine (refer to <i>Section 3- Installation</i>).
2	Invoke AUDIT to collect the TPN data as described in <i>Section 4.1 – Collecting TPN Data</i> .
3	Make the AUDIT data available to Sargent as described in <i>Section 4.1 – Making TPN Data Available to Sargent</i> .
4	If manual-entry data is available, refer to <i>Section 4.3- Manual Entry Data File</i> for details on its format and entry.
5	Invoke Sargent as described in <i>Section 5.1 – Invoking Sargent</i> .
6	Select as input the AUDIT report file you created and transferred as described in <i>Section 5.2 - Selecting Input Files</i> . If you are also using manual-entry data, select it as described in <i>Section 5.2 - Selecting Input Files</i> .
7	Specify an output file and format as described in <i>Section 5.3 - Specifying Output and Log Files</i> .
8	Generate the report as described in <i>Section 5.4 – Generating Reports</i> .

6.3 TPS-Only Report

If only the TPS network is to be audited, perform the procedure in Table 12.

Table 12 - Procedure for Auditing the TPS Network

Step	Action
1	Ensure that VMT-Net is installed on all TPS nodes that are to be audited, and that Sargent is installed on an Microsoft Windows machine. Installation is described in <i>Section 3 - Installation</i> .
2	Invoke VMT-Net to collect the TPS data as described in <i>Section 4.2 – Collecting TPS Data</i> . Ensure that you specify all nodes to be audited in the Select Computers input area.
3	Invoke Sargent as described in <i>Section 5.1 – Invoking Sargent</i> .
4	Select as input the VMT-Net report file you created, as described in <i>Section 5.2 - Selecting Input Files</i> .
5	Specify an output file and format as described in <i>Section 5.3 - Specifying Output and Log Files</i> .
6	Generate the report as described in <i>Section 5.4 – Generating Reports</i> .

6.4 TPN and TPS Report

If both the TPN and TPS networks are to be audited, perform the procedure in Table 13

Table 13 - Procedure for Auditing TPN and TPS Networks

Step	Action
1	Ensure that: • AUDIT is installed on a GUS, a US, or a UxS on the TPN network • File Transfer is installed to make the AUDIT data available to Sargent • VMT-Net is installed on all TPS nodes that are to be audited • Sargent is installed on an Microsoft Windows machine. Installation is covered in <i>Section 3 - Installation..</i>
2	Invoke AUDIT to collect the TPN data as described in <i>Section 4.1 – Collecting TPN Data.</i>
3	Make the AUDIT data available to Sargent as described in <i>Section 4.1 – Making TPN Data Available to Sargent.</i>
4	If manual-entry data is available, consult <i>Section 4.3- Manual Entry Data File</i> for details on its format and entry.
5	Invoke VMT-Net to collect the TPS data as described in <i>Section 4.2 – Collecting TPS Data.</i> Ensure that you specify all nodes to be audited in the Select Computers input area.
6	Invoke Sargent as described in <i>Section 5.1 – Invoking Sargent.</i>
7	Select as input the AUDIT report file you created and transferred in Steps 2 and 3, and the VMT-Net report file you created in Step 5, as described in <i>Section 5.2 - Selecting Input Files.</i> If you are also using manual-entry data, select it as described in <i>Section 5.2 - Selecting Input Files.</i>
7	Specify an output file and format as described in <i>Section 5.3 - Specifying Output and Log Files.</i>
8	Generate the report as described in <i>Section 5.4 – Generating Reports.</i>

6.5 Performing Further Analysis of Sargent Report Data

Sargent's CSV-format output may be read by other external applications and used for further analysis. This scenario presents one simple example of such analysis, using the Microsoft applications Excel and Access to prepare LCN data for simple queries.

Table 14 - Procedure for Further Analysis of Sargent Report Data

Step	Action
1	Generate a Sargent report in CSV format as described in this document. For the purposes of this example, use the filename scenario-6.4.csv
2	Invoke MS Excel, and use File > Open to load scenario-6.4.csv
3	Delete the rows up to <i>and including</i> the column headers for the software list. In this example, this is rows 1 through 659.
4	Delete all rows <i>following</i> the software list; In this example, that would be rows 239 through 537.
5	Save the remaining data as a new CSV file. For this example, name the file as lcn-software.csv
6	Terminate Excel.
7	Invoke MS Access, and select the Open an existing file option. Browse to where you saved the lcn-software.csv file and open it.
8	Use the Access wizard to import the CSV data. The fields have been renamed from their default names of Field1 , Field2 , and so forth to Node Type , Software Name , Version , Revision , and Node Number for easier manipulation.
9	Now that the data is in the form of an Access table, you can perform queries and analyses. Answers to questions such as "How many different versions of GUS am I running?" or "Which nodes have not yet been upgraded to 640.3?" may be easier to obtain through such analysis than by visual inspection of the Sargent report, especially on very large networks.

6. Execution Scenarios for System Audit Tool

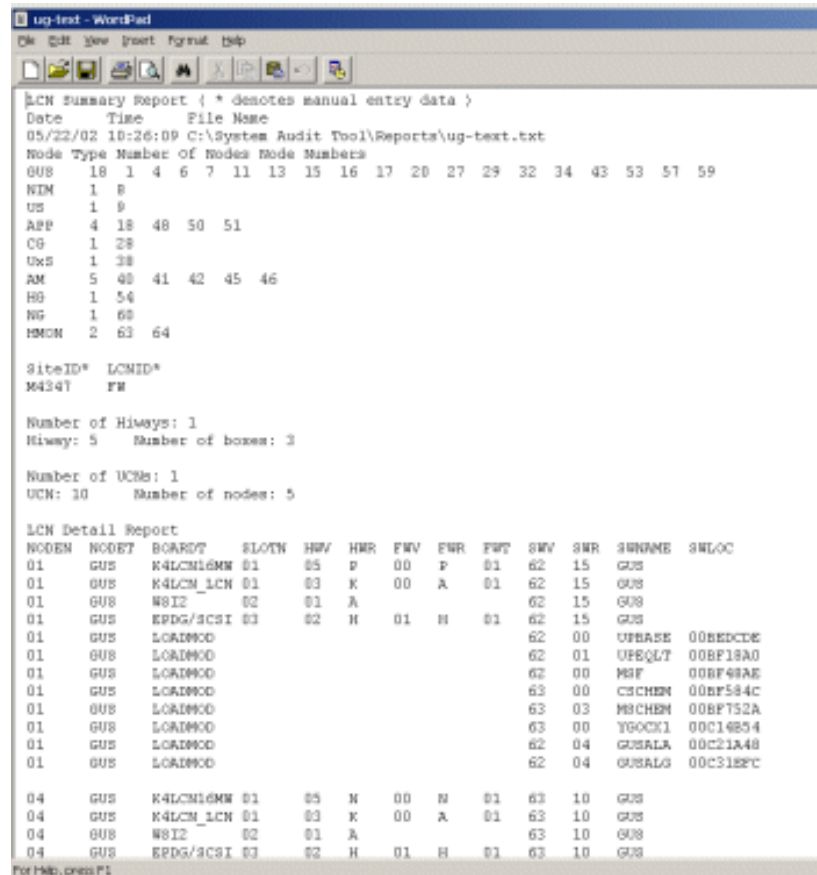
6.5. Performing Further Analysis of Sargent Report Data

7. File Formats and Post-Processing

7.1 Format of Text Output File

The text output report format is a simple flat-text file, readable by many different applications. Figure 15 shows Microsoft's WordPad application after loading one of Sargent's text-formatted report files. Only the first page of the full report appears in the figure.

Figure 15 - MS WordPad with Sargent Text-Format Report Loaded



```
ug-text - WordPad
File Edit View Insert Format Help

LCN Summary Report { * denotes manual entry data }
Date      Time      File Name
05/22/02 18:26:09 C:\System Audit Tool\Reports\ug-text.txt
Node Type Number of Nodes Node Numbers
GUS 18 1 4 6 7 11 13 15 16 17 20 27 29 32 34 43 53 57 59
NTM 1 8
US 1 9
APP 4 18 48 50 51
CG 1 28
UXS 1 38
AM 5 40 41 42 45 46
HG 1 54
NG 1 60
MCMN 2 63 64

SiteID* LCNID*
M4347 FM

Number of Hiways: 1
Hiway: 5 Number of boxes: 3

Number of UCNs: 1
UCN: 10 Number of nodes: 5

LCN Detail Report
NODEN NODET BOARDT SLOTH HWV HWR FWV FWR FWT SWV SWR SWNAME SWLOC
01 GUS K4LCM16MN 01 05 P 00 P 01 62 15 GUS
01 GUS K4LCM_LCN 01 03 K 00 A 01 62 15 GUS
01 GUS W8I2 02 01 A 62 15 GUS
01 GUS EPDG/SCSI 03 02 H 01 H 01 62 15 GUS
01 GUS LOADMOD 62 00 UPHASE 00BECDE
01 GUS LOADMOD 62 01 UFSQLT 00BF18A0
01 GUS LOADMOD 62 00 MBF 00BF48AE
01 GUS LOADMOD 63 00 CSCHEN 00BF584C
01 GUS LOADMOD 63 03 MBCHEN 00BF752A
01 GUS LOADMOD 63 00 YGCKN1 00C14854
01 GUS LOADMOD 62 04 GUSALA 00C21A48
01 GUS LOADMOD 62 04 GUSALG 00C31BFC

04 GUS K4LCM16MN 01 05 N 00 N 01 63 10 GUS
04 GUS K4LCM_LCN 01 03 K 00 A 01 63 10 GUS
04 GUS W8I2 02 01 A 63 10 GUS
04 GUS EPDG/SCSI 03 02 H 01 H 01 63 10 GUS

For Help, press F1
```

7. File Formats and Post-Processing

7.1. Format of Text Output File

The following sections describe the various reports contained within the file, and describe the data in each column within those reports. In every report, the presence of an asterisk (“*”) in a column or field name indicates that the column or field’s data was entered manually rather than collected automatically by AUDIT or VMT-Net.

LCN Summary Report

This report provides an overview of the LCN, Data Hiways, and UCNs on the audited TPN network. The items in the report are defined in Table 15.

Table 15 - Definitions of LCN Summary Report Items

Column/Field Name	Data provided
Node Type	Name of the node type (e.g., GUS, AM, HG, etc.)
Number of Nodes	Number of nodes of the given type
Node Numbers	List of node numbers for the nodes of the given type.
SiteID*	ID of the site (manual entry on AUDIT command line)
LCNID*	Name of this LCN (manual entry on AUDIT command line)
TPNVer*	Version of TPN software release (present if the tpn-rev manual entry directive was specified)
Number of Hiways	Number of Data Hiways on the audited TPN network
Hiway	Hiway number for each Data Hiway
Number of boxes	Number of hiway boxes for each Data Hiway
Number of UCNs	Number of UCNs on the audited TPN network
UCN	UCN number for each UCN
Number of nodes	Number of UCN nodes for each UCN

LCN Detail Report

This report provides a detailed listing of the individual boards and load modules in each LCN node. The items in the report are defined in Table 16.

Table 16 - Definitions of Items in LCN Detail Report

Column Name	Data provided
NODEN	Number of the node containing this board.
NODET	Name of the node type (e.g., GUS, AM, HG, etc.)
BOARDT	Board type, for boards LOADMOD for load modules
SLOTN	Board slot number
HWV	Hardware version of the board
HWR	Hardware revision of the board
FWV	Firmware version of the board
FWR	Firmware revision of the board
FWT	Firmware type of the board
SWV	Personality version, for LCN boards Load module version, for load modules
SWR	Personality revision, for LCN boards Load module revision, for load modules
SWNAME	Name of the personality software or load module
SWLOC	Blank for LCN boards Load address, for load modules

7. File Formats and Post-Processing

7.1. Format of Text Output File

Hiway Detail Report

This report provides a detailed listing of the individual cards in Data Hiway boxes for each Hiway active on the LCN. The items in the report are defined in Table 17.

Table 17 - Definitions of Items in Hiway Detail Report

Column Name	Data provided
HWY	Data Hiway number
BOXN	Number of the Hiway box containing this card
BOXT	Hiway box type
CARDT	Card type
CARDN	Card number
RV1*	First card revision (from manual-entry data file); blank if none given
RV2*	Second card revision (from manual-entry data file); blank if none given
BOXSIZE	EXTENDED if the box is extended, blank if not

UCN Detail Report

This report provides a detailed listing of the individual cards in UCN nodes for each UCN active on the LCN. The items in the report are defined in Table 18.

Table 18 - Definitions of Items in UCN Detail Report

Column Name	Data provided
UCN	UCN number
NODEN	Number of the UCN node containing this card
NODET	UCN node type
CARDT	Card type
MODN	The module number, for IOPs - the suffix "a" denotes a single card, or the primary card of a redundant pair - the suffix "b" denotes the secondary card of a redundant pair Blank for non-IOP cards
FILEN	Number of the card file containing this card
CARDN	Logical number of this card within the UCN node
FILEP	Position of this card within its card file
HWR	Card's Hardware revision of the card.
FWR	Firmware revision of the card
SWV	Software version loaded on this card; blank if none
SWR	Software revision loaded on this card; blank if none
SWNAME	Name of the software loaded on this card; blank if none
REDUNCY	Redundancy package option
FTATYPE*	Type (model number) of the FTA associated with this IOP (from manual entry data); blank if none given
REV*	Revision of the FTA associated with this IOP (from manual entry data); blank if none given

7. File Formats and Post-Processing

7.1. Format of Text Output File

Board List

This report provides a detailed listing of all boards/cards present on the LCN being audited, and on all the Data Hiways and UCNs connected to it. It contains some of the same data as the other three detailed reports described above, but in this report the data is grouped by board/card type. The items in the report are defined in Table 19.

Table 19 - Definitions of Items in Board List

Column Name	Data provided
NODET	LCN for boards in LCN nodes Type of Hiway box containing this card, for cards in Data Hiway boxes Type of UCN node containing this card, for cards in UCN nodes
BOARDT	Board or card type.
HWV	Hardware version of the board or card (if applicable)
HWR	Hardware revision of the board or card (if applicable)
FWV	Firmware version of the board or card (if applicable)
FWR	Firmware revision of the board or card (if applicable).
FWT	Firmware type of the board or card (if applicable)
LCN NODE/HIWAY BOX/UCN NODE	For boards in LCN nodes, number of the LCN node containing this board. For cards in Data Hiway boxes, contains two columns • Number of the Data Hiway to which the box containing this card is connected • Number of the box which contains this card For cards in UCN nodes, contains two columns • Number of the UCN to which the node containing this card is connected • Number of the node which contains this card

Software List

As with the Board List, this report presents some of the same data contained in the LCN, and UCN detailed reports described above, but ordered in a different way: This report is grouped and sorted by the name of the software loaded on a particular LCN or UCN node. The items in the report are defined in Table 20.

Table 20 - Definitions of Items in Software List

Column Name	Data provided
NODET	LCN for software loaded on LCN nodes For software loaded on UCN nodes, type of UCN node containing this software
SWNAME	For LCN software, name of the LCN personality or load module, For UCN software, the name of the UCN software component.
SWV	Software version.
SWR	Software revision.
LCN NODE/UCN NODE	For LCN software, number of the LCN node in which the software is loaded For UCN software, contains two columns: • Number of the UCN to which the node containing this software is connected • Number of the UCN node which contains this software

7. File Formats and Post-Processing

7.1. Format of Text Output File

TPS Summary

This report provides a summary of TPS software packages installed on each audited TPS node. One report is present for each node. The items in the report are defined in Table 21.

Table 21 - Definitions of Items in TPS Summary

Column/Field Name	Data Provided
Computer Name	Name of the computer whose audit data follows.
Report taken on:	Time/date the VMT-Net report was generated for this node.
Last Updated on:	The last time this computer was scanned by VMT-Net
Node status	Install status of the product installed on this node: • Blank for no conflict • ! (exclamation mark) indicates that a conflict exists with this component • X indicates that this component is missing If VMT-Net was asked to scan for all releases instead of just the most recent, then there may be several of these product/package groups, one for each release of the product that was detected
Node name/version	• Name of the product loaded on this node • Version number of the product, or blank if none
Package status	Install status of this individual package: • Blank for no conflict • ! (exclamation mark) indicates that a conflict exists with this component • An "X" indicates that this component is missing One package line will appear for each package detected.
Package name/version	• Name of the package • Version number of the package, or blank if none

7.2 Format of CSV Output File

The format of the CSV output file is identical to the format of the text output file as described in *Section 7.1 – Format of Text Output File*, except it conforms to the CSV file format. That is, it has commas separating the columns/fields instead of spaces. Other than this difference, the layout of the sections and the meaning of the columns within the sections are the same as for the text output file.

Figure 16 shows Microsoft's Excel application after loading one of Sargent's CSV-formatted report files. Only the first page of the full report appears in the figure.

Figure 16 - MS Excel with Sargent CSV-Format Report Loaded

Microsoft Excel - ug-csv

File Edit View Insert Format Tools Data Window Help

LCN Summary Report (* denotes manual entry data)

A	B	C	D	E	F	G	H	I	J	K	L	M	N
LCN Summary Report (* denotes manual entry data)													
Date	Time	File Name											
5/22/2002	10:26:09	C:\System Audit Tool\Reports\ug-csv.csv											
Node Type Number of Node Numbers													
GUS	18	1	4	6	7	11	13	15	16	17	20	27	29
NIM	1	8											
US	1	9											
APP	4	18	48	50	51								
CG	1	28											
UxG	1	38											
AM	5	40	41	42	45	46							
HG	1	54											
NG	1	60											
HMON	2	63	64										
SiteID*	LCNID*												
M4347	FW												
Number of Hiways: 1													
Hiway: 5	☐ Number of boxes: 3												
Number of UCNs: 1													
UCN: 10	☐ Number of nodes: 5												
LCN Detail Report													
NODEN	NODET	BOARDT	SLOTN	HWV	HWR	FWY	FWR	FWT	SWV	SWR	SWNAME	SWLOC	
1	GUS	K4LCN16MW	1	5	P	0	P	1	62	15	GUS		
1	GUS	K4LCN_LCN	1	3	K	0	A	1	62	15	GUS		
1	GUS	WSI2	2	1	A				62	15	GUS		
1	GUS	EPDG/SCSI	3	2	H	1	H	1	62	15	GUS		
1	GUS	LOADMOD							62	0	UPRASE	00RFXCDF	

Ready

7.3 Suggestions for Programmatic Use

The .CSV output format created by Sargent may be read by numerous applications, including Microsoft's Excel spreadsheet and Access database. This allows the data compiled by Sargent to be used in further analysis and comparison operations, as determined by the user. Such operations are clearly rather open-ended; precisely which operations should be performed on which data is dependent on what sort of results the user desires. Refer to *Section 6.5* for one example of how such processing might be performed.

8. Remote Operation

8.1 Types of Remote Operation

The System Audit Tool and its external components may be operated remotely from a computer in a different location than the one actually performing the processing. Two common ways of doing this are by using the CarbonCopy application or the Desktop Streaming application.

After the remote connection is established, the AUDIT application may be invoked using the GUS Native Window. Other Microsoft Windows operations can be performed as permitted by these remoting applications.

8.2 Remote Operation Using CarbonCopy

For remote operation using CarbonCopy, the CarbonCopy application must be installed on both the local and remote machines. (CarbonCopy is an optional application distributed as part of TPS.) For installation and use of CarbonCopy, please load the user guide from <http://www.altiris.com/products/carboncopysol>.

8.3 Remote Operation Using Desktop Streaming

For remote operation using Desktop Streaming, a small executable must be downloaded onto the target machine. For use of Desktop Streaming, please refer to the Honeywell Solution Support Online website at <http://www.ssol.acs.honeywell.com/media/desktop/index.html>

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